# Precision heating and cooling executive brief

THERMA COMMONS | November 2, 2026 | Version 1.0

## The proposal

THERMA COMMONS is a proposed modular appliance platform that heats, actively cools and verifies temperature for defined food and beverage loads. We seek $25,000 for a 12-week feasibility program, with an optional $10,000 in-kind contribution. The program will establish whether a cleanable stirred-vessel demonstrator can reach a selected temperature reproducibly and with practical time and energy use.

## The problem and insight

Users commonly specify cooking time while actually wanting a temperature outcome. Heating can leave uneven temperatures; cooling often requires waiting, transfer or improvised handling. The useful opportunity is verified control of the full thermal journey. A surface reading or precise display alone does not establish internal food temperature.

## The initial solution

A removable food-compatible vessel sits in a heating and cooling dock. Mixing reduces gradients. A control sensor and an independent reference measurement establish endpoint performance. The controller approaches a selected target, checks stability and reports measured versus unverified status. The first experiments use water and defined simulants; raw-food cooking and microbiological claims are excluded.

## The larger platform

After feasibility, the architecture extends to shallow meal trays, paired cooking and chilling chambers, accessible controls, heat recovery when useful, reusable thermal storage, repairable modules and community-kitchen variants. Solid-state RF and unconventional refrigeration remain optional research tracks. Bespoke luxury finishes and quieter packaging follow functional proof.

## Twelve-week evidence deliverables

Requirements and hazard review; reference-calibrated bench demonstrator; a randomized validation dataset; endpoint, gradient, time and wall-energy results; cleaning assessment; failure-injection results; an independent review; and a costed next-stage decision. Primary target: at least 95 percent of final blinded water trials within +/-1 degree C of the selected target under a declared operating envelope. This is a development target, not achieved performance or a food-safety guarantee.

## What a partner receives

A defined statement of work, milestone reports, agreed attribution, a demonstration and access to agreed nonconfidential results. Intellectual property, publication and licensing rights are negotiated separately. Sponsorship does not automatically grant equity, exclusivity, certification or patent ownership.

## Why this approach can move faster

Start with one measurable task using established heating and refrigeration components. Spend first on measurement, controls and cleanability. Release funding in stages. Expand only when evidence supports the next level of complexity.

## Project contact

Gerard Ando | Project lead | g.ando@gkitchen.in. Project status: concept-stage; specialist engineering collaborators sought. The working name is subject to trademark clearance.